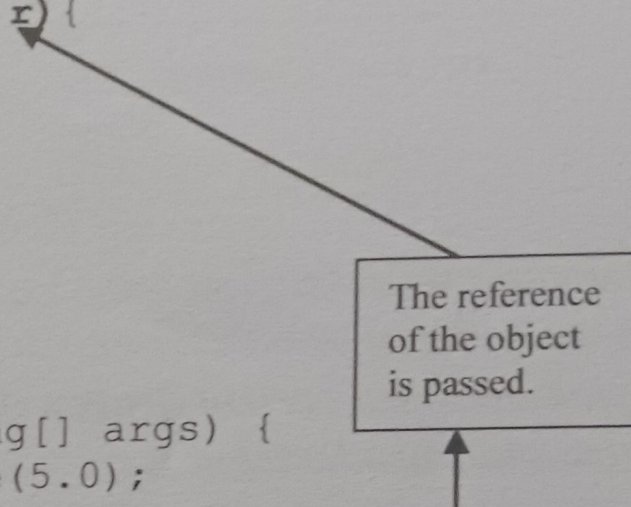


- 2.48 The following program shows the example of Pass-by-reference.
[Contoh atur cara berikut menunjukkan contoh bagi Penghantaran-secara-Rujukan]

```
1 class Circle {
2     private double radius, area ;
3     public Circle(double r){
4         radius=r;
5     }
6     public void findAreas(Circle r) {
7         area= radius*2.14*2.14;
8     }
9     public double getArea() {
10        return area;
11    }
12 }
13
14 public class PassByRef {
15     public static void main(String[] args) {
16         Circle circle1 = new Circle(5.0);
17         circle1.findAreas(circle1);
18         System.out.println ("The area is "+ circle1.getArea());
19     }
20 }
```



The reference of the object is passed.

- 2.49 Returning an object means that an object is returned as the return value.
[Pemulangan suatu objek bermakna suatu objek dipulangkan sebagai suatu nilai pulangan.]

```
1 class Circle {
2     private double radius;
3     public Circle(double r){
4         radius=r;
5     }
6
7     public Circle incrRadius(Circle r, Circle s){
8         Circle a = new Circle(0.0);
9         a.radius = r.radius+s.radius;
10        return a;
11    }
12 }
13
14 public class RetObject {
15     public static void main(String[] args){
16         Circle circle1 = new Circle(5.0);
17         Circle circle2 = new Circle(6.0);
18         Circle circle3 = new Circle(0.0);
19         circle3=circle2.incrRadius(circle1,circle2);
20     }
```

- 2.50 Java provide a standard practice to name a method

Trace the following program code and show the output. Draw a memory layout for the program.

[Jejak aturcara berikut dengan dan tunjukkan outputnya. Lukis hamparan ingatan untuk aturcara tersebut.]

```
1 public class ParameterPassing {
2     public static void main (String[] args) {
3         ParameterTester tester = new ParameterTester();
4         int a1 = 111;
5         Num a2 = new Num(222);
6         Num a3 = new Num(333);
7
8         System.out.println ("Before calling change:");
9         System.out.println ("a1\ta2\ta3");
10        System.out.println (a1 + "\t" + a2 + "\t" + a3 + "\n");
11
12        tester.changeValues(a1, a2, a3);
13
14        System.out.println ("After calling change values:");
15        System.out.println ("a1\ta2\ta3");
16        System.out.println (a1 + "\t" + a2 + "\t" + a3 + "\n");
17    }
18 }
19
20 class Num {
21     private int value;
22     public Num (int update) {
23         value = update;
24     }
25
26     public void setValue(int update) {
27         value = update;
28     }
29
30     public String toString() {
31         return value + " ";
32     }
33 }
34
35 class ParameterTester {
36     public void changeValues (int f1, Num f2, Num f3) {
37         System.out.println ("Before changing the values:");
38         System.out.println ("f1\tf2\tf3");
```



```

39      System.out.println (f1+"\t"+f2+"\t"+f3+"\n");
40
41      f1 = 999;
42      f2.setValue(888);
43      f3 = new Num(777);
44
45      System.out.println ("After changing the values:");
46      System.out.println ("f1\tf2\tf3");
47      System.out.println (f1+"\t"+f2+"\t"+f3+"\n");
48  }
49  }
50

```

2. Identify errors in the Program Trial below. You are required to give a brief explanation on each of the identified error and fix the error.

[Kenalpasti ralat-ralat yang terdapat di dalam Aturcara Trial. Kamu dikehendaki menerangkan secara ringkas ralat-ralat yang dikenalpasti dan betulkan ralat-ralat tersebut.]

```

1  //Program 2.7
2  class Trial
3  {
4      private int a;
5      private String b;
6
7      Trial(int _a)
8      {
9          a=_a;
10         b=_b;
11     }
12
13     public void setAB(int num, String str)
14     {
15         a=num;
16         b=str;
17     }
18     private int getA()
19     {
20         return a;
21     }
22
23     public void show()
24     {
25         System.out.println("A: "+getA());
26         System.out.println("B: "+b);
27     }
28 }
29
30 class TrialTest
31 {
32     public static void main(String[]a)
33     {
34         Trial ac=new Trial(5,"Five");
35         ac.a=10;
36         ac.b="Ten";
37         ac.setAB(6,"Six");
38         getA();
39         ac.show();

```



```
show();  
ac.getA();
```

```
}
```

```
}
```

SE 6: CONSTRUCTOR AND METHOD OVERLOADING

N 6: *KONSTRUKTOR DAN KAEDAH SARAT*

T [*KONSEP*]:

Within a constructor or a method, the variable names must be unique. There should be no two local variables or instance variables with the same name in the same scope.

dalam sesuatu konstruktor atau kaedah, nama-nama pemboleh
dalam skop yang sama.